

Very Important Points to Remember



Electronic Configuration

The Aufbau Principle

According to the **Aufbau Principle**, electrons fill subshells in order of increasing energy levels. The specific order in which the orbitals fill up is as follows:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p, and so on.

Orbital Capacities

Each orbital type has a fixed maximum capacity for the number of electrons it can accommodate:

- **s orbitals:** 2 electrons
- **p orbitals:** 6 electrons
- **d orbitals:** 10 electrons
- **f orbitals:** 14 electrons

Determining Position in the Periodic Table

By using the concepts of electronic configuration, we can determine the specific location of an element:

1. Block of an Element

Once you have filled all the electrons, the **orbital in which the last electron is placed** represents the block (s, p, d, or f) where the element is located.

2. Period of an Element

To determine the period, look at the **highest principal quantum number (n)** of the valence (outermost) electrons. This number represents the period number of the element.

3. Group of an Element

The group number is determined based on the block:

- **(a) s-block:** The group number is equal to the **number of valence electrons**.
- **(b) p-block:** The group number is determined by the formula: **(number of valence electrons + 10)**.



Memorization Trick for Group Characteristics

- s-block (1 & 2) → metals, reactive, low melting
- p-block (13–18) → non-metals (except B, Al)
- d-block (3–12) → transition metals, colored, multiple oxidation
- Halogens (17) → reactive non-metals
- Noble gases (18) → inert

Property	Across a Period (Left → Right)	Down a Group (Top → Bottom)
Atomic Radius	Decreases ↓	Increases ↑
Ionization Energy	Increases ↑	Decreases ↓
Electronegativity	Increases ↑	Decreases ↓
Electron Affinity	Increases ↑	Decreases ↓
Metallic Character	Decreases ↓	Increases ↑
Shielding Effect	Constant	Increases ↑
Valence Electrons	Increase	Remain constant
Reactivity (Metals)	Decreases ↓	Increases ↑
Reactivity (Non-metals)	Increases ↑	Decreases ↓
Density	Generally, increases then decreases	Generally, increases ↑

🔑 Quick Logic to Remember

- **Across a period:** Nuclear charge increases → atoms hold electrons more strongly.
- **Down a group:** Number of shells increases → atoms become larger and electrons are held less tightly.

This is why:

- Metals become **more reactive down a group**.
- Non-metals become **more reactive across a period**.

✅ 2-second exam memory trick

- **Across period:** size ↓, attraction ↑
- **Down group:** size ↑, attraction ↓

This single rule explains most trends in the **Periodic Table**. 📘

Bond Type	How it Forms	Types of Atoms Involved	Common Examples
Ionic Bond	One atom transfers electrons to another atom forming positive and negative ions	Usually Metal + Non-metal	Sodium chloride (NaCl), Magnesium oxide (MgO), NaF, CaCl ₂
Covalent Bond	Atoms share electrons to complete their octet	Usually Non-metal + Non-metal	H ₂ , Cl ₂ , Carbon dioxide (CO ₂), Methane (CH ₄), NH ₃
Coordinate Covalent Bond	One atom donates both electrons of the shared pair	Lone-pair donor + electron acceptor	Ammonium ion (NH ₄ ⁺), Hydronium ion (H ₃ O ⁺), CO

If you see:

- Na, K, Mg, Ca, Al with non-metals → Ionic
- C, H, O, N, Cl together → Covalent
- NH₄⁺, H₃O⁺, CO → Coordinate bond present

All Examples of Ionic Bonds in the Book

NaCl, KCl, NaF, NaBr, MgF₂, MgO, AlCl₃,

All Examples of Ionic Bonds in the Book

H₂, F₂, O₂, N₂, Cl₂, CH₄, H₂O, NH₃, HCl, CH₃OH, CO₂, HCN, CS₂, C₂H₄

Non-Polar Covalent Molecules

H₂, F₂, O₂, N₂, Cl₂, CH₄, CO₂, CS₂, C₂H₄

Polar Covalent Molecules

H₂O, NH₃, HCl, CH₃OH, HCN

Quick Exam Trick

- Same atoms → always non-polar
- Bent or pyramidal molecules → usually polar
- Linear symmetrical molecules → non-polar



Famous Examples of Coordinate Covalent Bonds

Molecule / Ion	Lone Pair Donor	Atom Accepting Electrons
Ammonium ion (NH_4^+)	Nitrogen (N)	Hydrogen ion (H^+)
Hydronium ion (H_3O^+)	Oxygen (O)	Hydrogen ion (H^+)
Carbon monoxide (CO)	Oxygen (O)	Carbon (C)
Sulfur dioxide (SO_2)	Oxygen (O)	Sulfur (S)
Ammonia–boron trifluoride complex (NH_3BF_3)	Nitrogen (N)	Boron (B)

2 Types of Intermolecular Forces

A) Dipole–Dipole Forces

Definition:

- Attraction between the **positive end of one polar molecule** and the **negative end of another polar molecule**.

Why They Exist:

- Polar molecules have **partial charges** due to **unequal sharing of electrons**.
- Opposite charges attract.

Example:

- HCl (Hydrogen of one HCl molecule is δ^+ , Chlorine is δ^-) → molecules attract each other.

Another Example:

- HBr, SO_2 , CH_3Cl



B) Hydrogen Bonding

Definition:

- Special type of **dipole–dipole attraction** where **H is covalently bonded to highly electronegative atoms like F, O, or N**, and H interacts with a lone pair on another electronegative atom.

Why They Exist:

- H has **partial positive charge (δ^+)** because electronegative atom pulls electron density.
- Lone pair on nearby F, O, or N attracts H, forming a **hydrogen bond**.

Example:

- Water (H_2O): $\text{O}-\text{H}\cdots\text{O}$
- NH_3 : $\text{H}-\text{N}\cdots\text{N}$
- HF : $\text{H}-\text{F}\cdots\text{F}$

Hydrogen bonding explains:

- High boiling point of water (100°C) vs H_2S (-60°C)
- Ice being less dense than water
- NH_3 being more soluble in water than PH_3

3 Why Intermolecular Forces Exist if Covalent Bonds Already Exist?

Important Point:

- Covalent bonds are **inside the molecule** → hold atoms together.
- Intermolecular forces are **between molecules** → hold molecules together in liquid or solid phase.

Example: Water (H_2O)

1 Covalent bonds: $\text{H}-\text{O}-\text{H}$ → keeps molecule intact

2 Hydrogen bonds: $\text{H}\cdots\text{O}$ between different H_2O molecules → gives water high boiling point, surface tension, ice structure

So, both are needed for full explanation of chemical and physical behavior.